

Validating **thecov**’s Gaussian covariance against DESI DR2 mocks: LRG1 and QSO, fibre assignment, bin width and the structure of the excess

Validation note, branch **thecov2**

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Abstract

We compare the windowed Gaussian covariance computed by **thecov** with the sample covariance of DESI DR2 mock power spectrum multipoles $P_{0,2,4}(k)$, for $0.02 \leq k \leq 0.3 \, h \, \text{Mpc}^{-1}$ in $\Delta k = 0.005 \, h \, \text{Mpc}^{-1}$ bins, using 859 holi v3 altmtl realisations (LRG1 and QSO) and 25 AbacusSummit realisations available both with and without fibre assignment (*complete* and *altmtl*). For QSO, **thecov** agrees with the mocks to $\lesssim 1\%$ in $\langle \chi^2 \rangle / n$. For LRG1 the mocks show $\langle \chi^2 \rangle / n \simeq 1.10$ (holi) and 1.10–1.17 (Abacus). Four results organise this excess. (i) Fibre assignment is not responsible: on identical Abacus realisations, altmtl and complete agree to $\Delta \langle \chi^2 \rangle / n = 0.000\text{--}0.023 \pm 0.014$. (ii) Changing the bin width separates a component of the excess that does not depend on Δk (“Gaussian-like”, $a \simeq 0.10$ for LRG1 P_0 and P_2 , $\simeq 0$ for QSO) from one proportional to Δk (non-Gaussian, b growing from 0.02 to 0.15 per $0.005 \, h \, \text{Mpc}^{-1}$ in P_0). (iii) Per-object weight scatter, $\langle w^2 \rangle \neq \langle w \rangle^2$, is handled correctly by a smoothed window $m(\mathbf{x})$; the naive window would have hidden part of the excess. (iv) The excess is concentrated in a few directions of data space: the leading one is a broad-band P_0 amplitude/dilation mode whose variance is $\simeq 6$ times **thecov**’s, the second a P_2 amplitude mode. With a few nuisance parameters marginalised, **thecov** underestimates the variance of an amplitude-like parameter by 15–25% and that of a quadrupole amplitude by $\simeq 70\%$ for LRG1 at $k_{\text{max}} = 0.2 \, h \, \text{Mpc}^{-1}$ (more at 0.3). The per-mock estimator normalisation explains only $\simeq 2\%$ of the amplitude scatter, so the isotropic local-average effect is small; the amplitude modes point to super-sample (including tidal) and connected non-Gaussian terms that **thecov** does not contain.

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1 Set-up

Mocks and spectra. Power spectrum multipoles are those of the DESI DR2 full-shape pipeline (`jaxpower`), read from the `summary_statistics/full_shape/base` directory for three mock families:

- **holi v3 altml:** 859 realisations, LRG1 ($0.4 < z < 0.6$) and QSO ($0.8 < z < 2.1$), NGC, SGC and GCcomb;
- **AbacusSummit 2nd gen. DR2 complete** and **altmtl:** the same 25 realisations without and with fibre assignment, LRG1.

We use $0.02 \leq k \leq 0.3 h \text{ Mpc}^{-1}$ in $\Delta k = 0.005 h \text{ Mpc}^{-1}$ bins ($n = 3 \times 56 = 168$ data points), and the coarser binnings $\Delta k = 0.01$ and $0.02 h \text{ Mpc}^{-1}$ (Sec. 6). The window (randoms, weights) is taken from one realisation per family (holi mock 173, Abacus mock 0), with one random file.

Covariance. `thecov` computes the windowed Gaussian covariance in configuration space from pair counts of randoms carrying the clustering window $W = m^2$ and the shot-noise window $S = \bar{n}\langle w^2 \rangle$, where

$$m(\mathbf{x}) = \bar{n}(\mathbf{x}) \langle w \rangle(\mathbf{x}) = \text{E} \left[\sum_g w_g \delta_D(\mathbf{x} - \mathbf{x}_g) \right] \quad (1)$$

is the mean weighted density, $w = \text{WEIGHT} \times \text{WEIGHT_FKP}$. Three constructions of m at each random are compared: *smoothed* ($\alpha \rho_r(z)$ times the mean weight of the 32 nearest randoms; the default), *nx* ($\text{NX} \times \text{WEIGHT_FKP}$, no smoothing), and *naive* (each random's own weight). The model power spectrum is the mock mean rescaled by $\text{norm}/\int m^2$ (Sec. 2). GCcomb is the norm-weighted mean of NGC and SGC in the files (to 5×10^{-14}), so its covariance is the corresponding combination of the cap covariances.

Statistics. With N mocks, $\langle \chi^2 \rangle/n$ is the mean of $(\hat{\mathbf{P}}_i - \bar{\mathbf{P}})^T C^{-1} (\hat{\mathbf{P}}_i - \bar{\mathbf{P}}) / [n(1 - 1/N)]$, expected to be $1 \pm \sqrt{2/(nN)}$ (± 0.004 for holi, ± 0.022 for Abacus). We also use the per-element variance ratio $\sigma_{\text{mock}}^2 / \sigma_{\text{thecov}}^2$, the eigenvalues of the whitened mock covariance $E = C^{-1/2} \hat{C} C^{-1/2}$ against the Marchenko–Pastur (MP) range, and variances along chosen directions.

2 Corrections required to compare with the DESI files

Before any physics, four conventions had to be matched. Together they moved the variance ratio from 0.77 to $\simeq 1.1$.

1. **Model normalisation.** `jaxpower` normalises by $\text{norm} = \alpha \sum_c D_c R_c / V_c$ on $10 h^{-1} \text{ Mpc}$ cells, which smooths over the fine veto masks. Its mean $\hat{P}$ is therefore $(\int m^2 / \text{norm}) \times$ the windowed P . We find $\int m^2 / \text{norm} = 1.22$ (LRG1, holi), 1.11 (QSO), 1.15–1.18 (LRG1, Abacus). Feeding the raw mock mean to `thecov` inflates the clustering terms by $(1.22)^2 \simeq 1.5$; the model must be $\bar{P} \times \text{norm} / \int m^2$.
2. **Shot noise.** The files subtract the realised shot noise, `num_shotnoise/norm`. Its catalogue estimate $\sum_d w^2 + \alpha^2 \sum_r w^2$ from a single random file is 16–19% too high (LRG1) because the spectra used many more randoms. We normalise the shot-noise window to the files' mean `num_shotnoise`.
3. **GCcomb** is combined from the caps exactly, as above.

4. $k_{\min} = 0.02 h \text{ Mpc}^{-1}$. The first bins ($kL \lesssim 1$) are outside the validity of any local windowed Gaussian covariance and make it non-positive-definite.

3 Summary of the comparison

mocks	bin	region	smoothed	nx	naive	$\Delta k \times 2$	$\Delta k \times 4$	$\pm$
holi	LRG1	NGC	1.096	1.112	1.031	1.120	1.204	0.004
holi	LRG1	SGC	1.099	1.114	1.067	1.123	1.190	0.004
holi	LRG1	GCcomb	1.098	1.110	–	1.123	1.199	0.004
holi	QSO	NGC	1.009	1.005	0.993	1.024	1.062	0.004
holi	QSO	SGC	1.000	1.001	0.990	1.013	1.043	0.004
holi	QSO	GCcomb	1.004	1.001	–	1.016	1.053	0.004
Ab. compl.	LRG1	NGC	1.127	1.128	1.119	1.100	1.184	0.022
Ab. compl.	LRG1	SGC	1.121	1.117	1.123	1.149	1.210	0.022
Ab. compl.	LRG1	GCcomb	1.174	1.173	–	1.184	1.268	0.022
Ab. altmtl	LRG1	NGC	1.104	1.131	1.054	1.082	1.120	0.022
Ab. altmtl	LRG1	SGC	1.121	1.092	1.144	1.114	1.175	0.022
Ab. altmtl	LRG1	GCcomb	1.157	1.166	–	1.162	1.211	0.022

Table 1: $\langle \chi^2 \rangle / n$ for $0.02 \leq k \leq 0.3 h \text{ Mpc}^{-1}$, $P_{0,2,4}$, with the three constructions of m (Sec. 1) at $\Delta k = 0.005 h \text{ Mpc}^{-1}$, and with the smoothed m at $\Delta k = 0.01$ and $0.02 h \text{ Mpc}^{-1}$. The last column is the statistical error at $\Delta k = 0.005$; it grows as $\sqrt{2}$ and 2 for the coarser binnings.

QSO is consistent with **thecov** at $\Delta k = 0.005$; at $\Delta k = 0.02$ it rises to 1.04–1.06, a non-Gaussian high- k tail (Sec. 6). LRG1 is not: $\langle \chi^2 \rangle / n = 1.10$ in holi, the same with nx and smoothed windows, and it grows with the bin width. QSO’s covariance is dominated by shot noise ($\bar{n}P < 1$), so it checks the shot-noise terms, the normalisation and the model correction; the LRG1 excess is in the clustering terms.

4 Weights: why m must be smoothed

The clustering terms of the Gaussian covariance contain two factors of m at (nearly) the same point. Pairs of randoms weighted by their own weights give $w_i w_j$, i.e. one factor of m at each end of the pair, which is the window of $\langle \hat{P} \rangle$, not of its covariance. Each end must carry an unbiased estimate of m^2 , and a single object cannot provide one: its own weight squared has expectation $\langle w^2 \rangle$, larger than $\langle w \rangle^2$ by $1 + \text{var}(w) / \langle w \rangle^2$. The second factor must come from neighbours, i.e. from a smoothed m . This is exact for stochastic weights that are independent between objects and of the density field: the clustering windows involve $\langle w \rangle$, the shot-noise windows $\langle w^2 \rangle$ (each random’s own w^2 estimates it without bias).

mocks	bin	region	$\langle w^2 \rangle / \langle w \rangle^2$	$\int m^2 / \text{norm}$		$\langle \chi^2 \rangle / n$	
				smoothed	naive	smoothed	naive
holi	LRG1	NGC	1.165	1.225	1.519	1.096	1.031
holi	LRG1	SGC	1.157	1.210	1.465	1.099	1.067
holi	QSO	NGC	1.039	1.110	1.218	1.009	0.993
holi	QSO	SGC	1.037	1.107	1.190	1.000	0.990
Ab. compl.	LRG1	NGC	1.000	1.177	1.179	1.127	1.119
Ab. compl.	LRG1	SGC	1.000	1.147	1.153	1.121	1.123

Table 2: Weight scatter of the data and its effect. The naive window overestimates $\int m^2$ by $\simeq \langle w^2 \rangle / \langle w \rangle^2$, as predicted, and inflates the covariance. In Abacus complete the weights barely scatter and the two windows coincide.

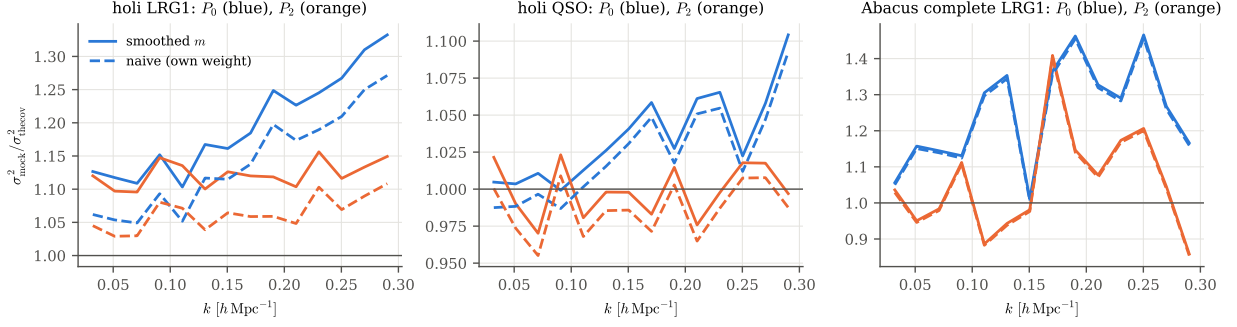


Figure 1: Variance ratio (mean over $0.02 \, h \, \text{Mpc}^{-1}$ -wide groups of bins, both caps) with the smoothed and naive windows. The naive window adds a nearly flat $\simeq 5$ –7% to `theov`’s LRG1 clustering variance and $\simeq 2\%$ to QSO’s. The weights in Abacus complete barely scatter and both constructions coincide (right), a null test.

Table 2 and Fig. 1 show the expected pattern: LRG1 weights scatter by 16% (completeness weights from fibre assignment), QSO by 4%, Abacus complete by 0.05%. The naive window increases $\int m^2$ by the weight scatter and the covariance accordingly. Note the trap: with the naive window, LRG1 looks *better* ($\langle \chi^2 \rangle / n = 1.03$ instead of 1.10), because the overestimated window partly compensates the missing terms discussed below. The control (Abacus complete, $\langle w^2 \rangle / \langle w \rangle^2 = 1.0005$) shows identical results for both, as it must.

5 Fibre assignment: Abacus altmtl versus complete

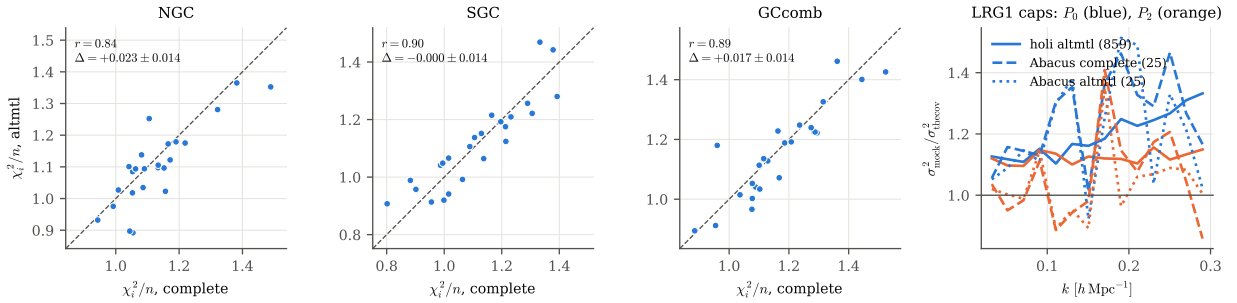


Figure 2: Left three panels: χ_i^2/n of each Abacus realisation with fibre assignment (altmtl) against the same realisation without (complete); r is the correlation coefficient and Δ the mean paired difference complete – altmtl. Right: variance ratio of P_0 (blue) and P_2 (orange) for holi altmtl (859 mocks) and the two Abacus versions (25 mocks).

Because the two Abacus versions share the underlying realisation, the per-mock χ_i^2 are strongly correlated ($r = 0.84$ – 0.90) and their difference is measured to ± 0.014 instead of the ± 0.03 of two independent sets. The differences, $+0.023 \pm 0.014$ (NGC), 0.000 ± 0.014 (SGC) and $+0.017 \pm 0.014$ (GCcomb), are consistent with zero, and the scale dependence of the variance ratio is the same with and without fibre assignment, and the same as in holi (Fig. 2, right). Fibre assignment adds at most $\simeq 2\%$ to $\langle \chi^2 \rangle / n$: the excess is not a consequence of the density-dependent selection or of the weights that compensate it, but is present in the complete mocks.

6 Bin width: Gaussian-like and non-Gaussian excess

The Gaussian covariance of a bin scales as $1/N_{\text{modes}} \propto 1/\Delta k$, while connected non-Gaussian terms (trispectrum, super-sample) are smooth functions of (k, k') and do not average down within a bin.

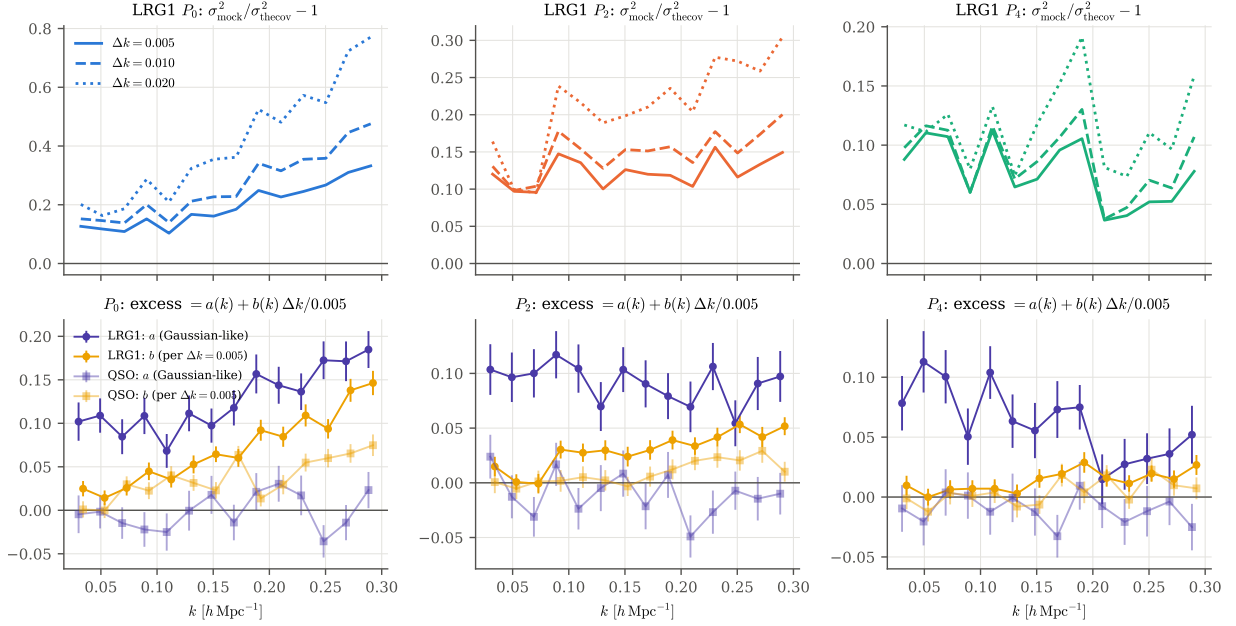


Figure 3: Top: LRG1 (holi, NGC and SGC averaged) excess variance for three bin widths. Bottom: the decomposition Eq. (2) for LRG1 (dark) and QSO (light): a (violet) does not depend on Δk , b (yellow) is proportional to it.

Writing the per-element excess at bin width $f \times 0.005 h \text{ Mpc}^{-1}$ as

$$\frac{\sigma_{\text{mock}}^2}{\sigma_{\text{thecov}}^2} - 1 = a(k) + b(k) f, \quad (2)$$

a collects anything with Gaussian scaling (a missing or mis-normalised Gaussian term) and b the non-Gaussian variance in units of the Gaussian variance of a $0.005 h \text{ Mpc}^{-1}$ bin. We fit Eq. (2) to $f = 1, 2, 4$ in $0.02 h \text{ Mpc}^{-1}$ -wide groups (errors from bootstrap over mocks).

Figure 3 shows:

- **Non-Gaussian part.** b grows with k and is largest in P_0 : from 0.02 at $k \simeq 0.03$ to 0.15 at $k \simeq 0.29 h \text{ Mpc}^{-1}$ for LRG1 (0.05 in P_2 , $\lesssim 0.03$ in P_4), and up to 0.07 in QSO P_0 . At $\Delta k = 0.02 h \text{ Mpc}^{-1}$ it is the dominant part of the excess at $k > 0.15 h \text{ Mpc}^{-1}$.
- **Gaussian-like part.** For QSO, $a \simeq 0$ at all k , as expected. For LRG1, however, $a \simeq 0.10$ in P_0 and P_2 over the whole range (rising to $\simeq 0.17$ in P_0 at $k > 0.2$), and 0.05–0.10 in P_4 . It is also present in the Abacus mocks (mean $a_0 \simeq 0.14\text{--}0.18 \pm 0.03$ in complete and altmtl). This corrects an earlier estimate ($a \simeq 0.025$) based on $\langle \chi^2 \rangle / n$ of the full vector, which mixes the two parts.

The Gaussian-like $\simeq 10\%$ is therefore specific to LRG1, independent of fibre assignment, and independent of the construction of m at the scales probed by the smoothed, nx and patch windows. Two candidates remain, both of which affect the clustering terms only (hence not QSO):

1. *Window structure below the smoothing scale.* An exact FFT-mesh Gaussian calculation with m taken directly from random counts in cells (window “B”), converged down to $1.5 h^{-1} \text{ Mpc}$, gave 6–9% more P_0 variance than **thecov**, and accounted for the low- k excess (mocks/B $\simeq 1.01\text{--}1.02$ at $k < 0.05$). With **thecov**’s own m the mesh calculation agrees with **thecov** to 0.3% (isotropic model). $W = m^2$ at a single point is itself a local approximation: near the many small veto holes and boundaries, $m(\mathbf{x})m(\mathbf{x} + \mathbf{r}) \neq m^2(\mathbf{x})$ for r within a correlation length.

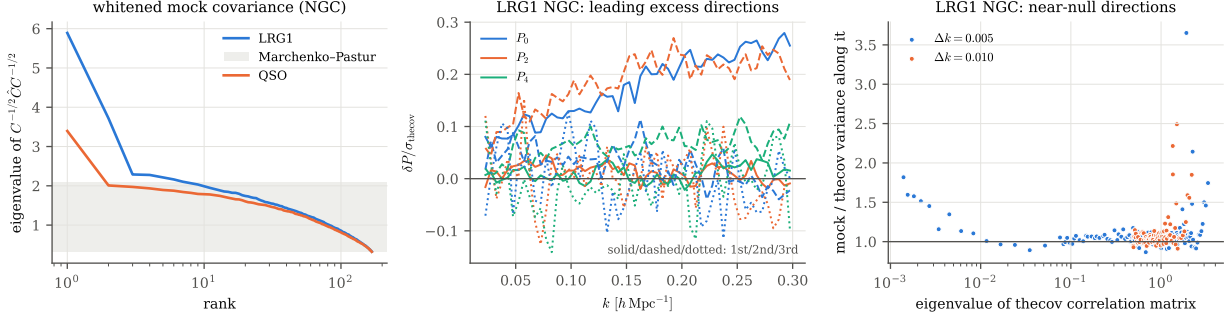


Figure 4: Left: eigenvalues of the whitened mock covariance $E = C^{-1/2}\hat{C}C^{-1/2}$ (holi NGC); grey: the MP range for $n = 168$, $N = 859$. Middle: the three leading excess directions for LRG1 NGC in data space, in units of `thecov`'s σ per element. Right: variance ratio along each eigenvector of `thecov`'s correlation matrix against its eigenvalue, at $\Delta k = 0.005$ and $0.01 h \text{Mpc}^{-1}$.

2. *The model normalisation.* The correction $\text{norm}/\int m^2$ is a single number ($1/1.22$) applied at all k . The relevant quantity is the window pair count at separations $\sim 10\text{--}100 h^{-1}\text{Mpc}$, which need not coincide with the $10 h^{-1}\text{Mpc}$ -cell norm; a 5% error in the effective model amplitude gives 10% in the clustering variance. LRG1 has the largest $\int m^2/\text{norm}$ of the samples considered, QSO the smallest.

Both can be tested with existing machinery: pair counts of $m \otimes m$ at $10\text{--}100 h^{-1}\text{Mpc}$ against norm and $\int m^2$ (item 2), and window B extended to $k = 0.3$ (item 1).

7 Directions in which the mocks and thecov differ

The excess is low-rank. If `thecov` were exact, the eigenvalues of E would fill the MP range. For QSO only one eigenvalue (3.4) lies above it. For LRG1, the top two eigenvalues are 5.9 and 3.7, and the rest of the spectrum is shifted up uniformly by $\simeq 10\%$, the Gaussian-like part a of Sec. 6 (Fig. 4, left).

What the leading directions are. Whitened templates $u = C^{-1/2}t$ identify them. The **first** direction (eigenvalue 5.9) contains 95% of the whitened $P_0+P_2+P_4$ amplitude template $t = \bar{P}$ and 91% of the dilation template $t = -k \partial \bar{P} / \partial k$; the two are nearly degenerate once whitened. It is a smooth, broad-band P_0 mode growing with k (Fig. 4, middle). The **second** (3.7) contains 73% of a P_2 -only amplitude template. Along the amplitude template itself, the mocks have 5.6 times `thecov`'s variance ($k \leq 0.3$); along the P_2 amplitude, 3.1 times.

Near-null directions at fine binning. At $\Delta k = 0.005 h \text{Mpc}^{-1}$, adjacent bins are strongly correlated by the window, and `thecov`'s correlation matrix has eigenvalues down to 10^{-3} . The corresponding eigenvectors are differences between neighbouring bins. Along them the mocks have 1.5–1.8 times the predicted variance; at $\Delta k = 0.01 h \text{Mpc}^{-1}$ the smallest eigenvalue is 0.5 and the ratio is $\simeq 1$ (Fig. 4, right). These directions depend on the detailed shape of the window kernel, a Gaussian-level effect, and a smooth non-Gaussian term cancels in them. They contribute a few per cent to $\langle \chi^2 \rangle / n$ at $\Delta k = 0.005$ and are irrelevant for smooth parameter derivatives, but a goodness-of-fit χ^2 will feel them.

8 Amplitude modes, super-sample covariance and the local-average effect

A long-wavelength mode δ_b larger than the survey changes the measured multipoles coherently: $\Delta \hat{P}_\ell(k) = [R_\ell(k) - 2P_\ell(k)] \delta_b$ for an estimator normalised by the realised mean density, where R_ℓ is

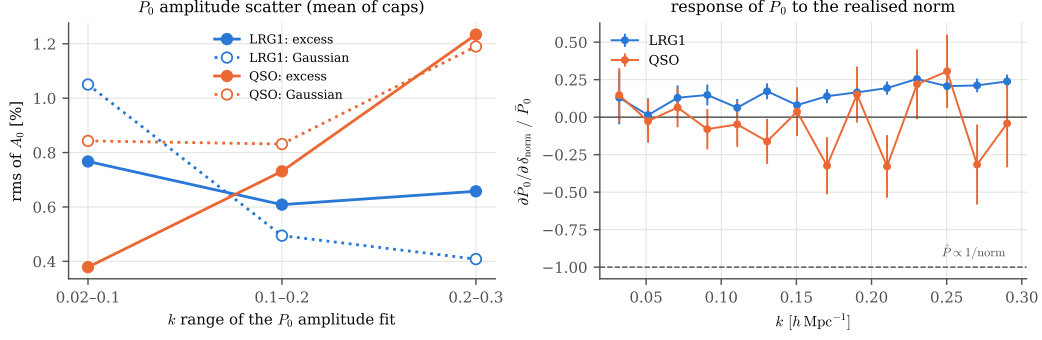


Figure 5: Left: rms of the fitted P_0 amplitude A_0 (template $\bar{P}_0$) over three k ranges: its excess over `thecov`'s prediction (solid) and the prediction itself (dotted), mean of the caps. Right: slope of $\hat{P}_0$ against the per-mock fluctuation of the estimator's normalisation, in units of $\bar{P}_0$, with standard regression errors.

the physical response (growth, dilation, bias) and $-2P_\ell$ the local-average (LA) effect. In redshift space the long mode's own anisotropy (its Kaiser term and tidal components) adds responses that change P_2 (and P_0) without changing the mean density (Akitsu, Takada & Li 2017; Wadekar & Scoccimarro 2020). These terms produce exactly the kind of low-rank amplitude-like excess seen in Sec. 7.

The k -dependence separates SSC from the trispectrum. We fit an amplitude to P_0 alone in three k ranges (Fig. 5, left). For LRG1 the excess amplitude scatter is nearly constant, 0.6–0.8% (NGC: 0.67, 0.51, 0.58%; SGC: 0.86, 0.70, 0.74%), present already at $k < 0.1 h \text{ Mpc}^{-1}$: the signature of a super-sample response. For QSO it grows from 0.3–0.4% at $k < 0.1$ to 1.0–1.5% at $k > 0.2 h \text{ Mpc}^{-1}$: the signature of the connected (and Poisson) trispectrum, with a weak super-sample part, consistent with QSO's much larger volume. The LRG1 P_2 amplitude has an excess of 3–5% in every range, and is only weakly correlated with the P_0 amplitude ($r \simeq 0.1$ –0.3): two independent long-mode responses, as expected for isotropic and tidal super-sample modes.

The local-average response, measured. `jaxpower`'s normalisation is computed for each mock from its own data and randoms, so the fluctuation $\delta_{\text{norm}} = \text{norm}_i / \langle \text{norm} \rangle - 1$ traces the survey-mean density of each realisation (plus Poisson noise). We find $\sigma(\delta_{\text{norm}}) = 0.43\%$ (LRG1 NGC) and 0.75% (SGC), against a Poisson expectation of $\simeq 0.13$ –0.26% and 0.18 –0.36%, so most of it is clustering. Regressing $\hat{P}_0(k)$ on δ_{norm} over the 859 mocks (Fig. 5, right):

- the slope is positive and small, $\partial \hat{P}_0 / \partial \delta_{\text{norm}} \simeq (0.1 \rightarrow 0.25) \bar{P}_0$ rising with k , far from the $-\bar{P}_0$ that a pure rescaling $\hat{P} \propto 1/\text{norm}$ would give. If the realised norm scales as $(1 + \delta_b)^2$ and $\hat{P}_0$ as $1 + (R - 2)\delta_b$, the slope is $(R - 2)/2$ diluted by the Poisson part: the physical response and the local-average effect nearly cancel for the isotropic mode, with a residual growing with k as the dilation term does;
- the correlation of the fitted amplitude with δ_{norm} is 0.14 (NGC) and 0.17 (SGC) for LRG1 and 0.00–0.02 for QSO. The survey-mean density therefore explains only $\simeq 2$ –3% of the LRG1 amplitude variance; removing it from each mock changes $\langle \chi^2 \rangle / n$ by 0.002–0.003 and the excess amplitude scatter from 0.54 to 0.54% (NGC).

The isotropic, density-tracing super-sample mode with its local-average counterpart is thus a minor contributor. The large, k -independent P_0 and P_2 amplitude scatter of LRG1 must come from long modes that do not change the survey's mean density (tidal and redshift-space anisotropic components of the background, and, if the mock randoms take their redshifts from each mock's own data, the radial modes they absorb) or from a broad-band component of the trispectrum. A

super-sample module for **thecov** must therefore include the anisotropic responses in the redshift-space formalism of Wadekar & Scoccimarro (2020) and model the local average for **jaxpower**’s realised normalisation; an isotropic σ_b^2 alone would underpredict it.

9 Impact on parameter errors

Since the excess is concentrated in amplitude-like directions, its impact on parameters depends on what is marginalised. We use four templates built from the mock mean: an overall amplitude A (template $\bar{P}_\ell$, a proxy for $b\sigma_8$), a quadrupole amplitude A_2 ($\bar{P}_2$ only, a proxy for $f\sigma_8$), dilation α ($-k \partial \bar{P}_\ell / \partial k$) and a constant in P_0 (shot noise). For each mock we fit them by generalised least squares with **thecov**’s C and compare the scatter of the estimates over the mocks with **thecov**’s prediction $(T^T C^{-1} T)^{-1}$. A ratio of 1 means **thecov**’s errors are right; the statistical error on each ratio is $\simeq 5\%$ for holi and $\simeq 30\%$ for Abacus.

mocks	bin	region	$k_{\max}$	one parameter			jointly			SN
				A	A_2	α	A	A_2	α	
holi	LRG1	NGC	0.2	3.03	2.17	2.90	1.15	1.73	0.97	0.93
holi	LRG1	NGC	0.3	5.65	3.13	5.51	1.33	2.24	1.15	1.28
holi	LRG1	SGC	0.2	2.84	2.08	2.72	1.25	1.76	1.17	1.17
holi	LRG1	SGC	0.3	4.75	2.52	4.56	1.68	1.91	1.53	1.49
holi	LRG1	GCcomb	0.2	3.02	2.10	2.91	1.15	1.72	1.02	1.05
holi	LRG1	GCcomb	0.3	5.54	2.89	5.42	1.39	2.07	1.22	1.32
holi	QSO	NGC	0.2	1.75	1.05	1.78	0.98	1.04	0.90	1.07
holi	QSO	NGC	0.3	2.52	1.05	2.66	1.01	1.04	0.97	1.42
holi	QSO	SGC	0.2	1.87	1.06	1.88	1.03	1.05	0.96	1.07
holi	QSO	SGC	0.3	2.52	1.00	2.62	1.09	1.01	1.05	1.39
holi	QSO	GCcomb	0.2	1.76	1.07	1.82	1.05	1.06	0.97	1.04
holi	QSO	GCcomb	0.3	2.47	1.03	2.64	1.09	1.01	1.08	1.43
Ab. compl.	LRG1	GCcomb	0.2	3.60	1.28	3.49	1.67	0.86	1.81	1.14
Ab. compl.	LRG1	GCcomb	0.3	3.70	2.04	3.15	2.19	1.49	1.50	1.93
Ab. altmtl	LRG1	GCcomb	0.2	3.79	1.17	3.58	1.56	0.80	1.56	1.11
Ab. altmtl	LRG1	GCcomb	0.3	3.27	1.81	2.66	2.22	1.26	1.13	1.43

Table 3: Variance of template parameters over the mocks divided by **thecov**’s prediction, fitting one parameter at a time and all four jointly.

Table 3 shows:

- Fitted alone, an amplitude or dilation parameter has 3 ($k_{\max} = 0.2$) to 5.6 ($k_{\max} = 0.3$) times **thecov**’s variance for LRG1, and 1.8–2.7 times for QSO. This is the leading excess direction of Sec. 7.
- With the four parameters fitted jointly, most of it is absorbed: for LRG1 at $k_{\max} = 0.2$ the variance ratios are 1.15–1.25 (A), 0.97–1.17 (α), 0.93–1.17 (SN), but 1.72–1.76 for the quadrupole amplitude A_2 . For QSO all joint ratios are within 10% of 1 except SN at $k_{\max} = 0.3$.
- The Abacus results are broadly consistent within their $\simeq 30\%$ errors, though systematically higher at $k_{\max} = 0.3$.

The quadrupole amplitude is the parameter most affected, since it is the second excess direction and is not degenerate with the nuisance parameters. These templates are a proxy for a full-shape fit, which marginalises over more nuisance parameters; the numbers should be redone with the derivatives of the actual DESI model, which is a direct extension of this test.

10 Robustness checks

The following checks, run earlier on the same data, support the statements above.

- **Accuracy of `thecov`.** On the real LRG1 NGC footprint, `thecov`'s P_0 variance agrees with an exact Gaussian calculation on an FFT mesh with the same m to 0.3% for an isotropic model (2–3% with the full model, a consequence of the mesh's global line of sight). The Monte Carlo noise of `thecov`'s pair counts is 1–2% at 2×10^9 near pairs; 10^{10} is recommended for final matrices.
- **Window constructions.** Smoothed and nx windows give $1/V_{\text{eff}}$ within 2% and $\langle\chi^2\rangle/n$ within 0.02 of each other at $\Delta k = 0.005$ (Table 1); a per-patch $n(z)$ (`nside` = 4–16) changes nothing, and the variation of $n(z)$ across the sky is at the Poisson level.
- **Cap independence.** The NGC and SGC amplitudes of the same realisation are uncorrelated ($r = -0.05$ for Abacus, -0.03 for holi), and GCcomb's amplitude scatter is reduced by $\simeq \sqrt{2}$, as expected for super-sample modes of two distant caps. This rules out an amplitude systematic shared by both caps.
- **Two mock families.** holi (approximate) and Abacus (N -body) give the same LRG1 excess and the same k -dependence, so it is not an artefact of the mock method.

11 Conclusions and recommendations

1. With the conventions of Sec. 2, `thecov` reproduces the QSO mock covariance to 1% and computes the Gaussian covariance of the DESI window to better than 1%.
2. The window must use a smoothed $m(\mathbf{x})$ (or $\text{NX} \times \text{WEIGHT_FKP}$, equivalent here), not the objects' own weights.
3. Fibre assignment does not contribute to the LRG1 excess at the 2% level in $\langle\chi^2\rangle/n$.
4. The LRG1 excess has three components: (a) a Gaussian-like $\simeq 10\%$ (unexplained; window structure below $\sim 10 h^{-1}\text{Mpc}$ and the model normalisation are the candidates, Sec. 6); (b) a non-Gaussian part growing with k , mostly in P_0 ; (c) two dominant amplitude modes (P_0 and P_2) of super-sample character, not traced by the survey-mean density.
5. For parameters, (c) matters most: a quadrupole-amplitude parameter has $\simeq 70\%$ more variance than `thecov` predicts even with nuisance parameters marginalised. Until non-Gaussian terms are added, a DR2 fit using `thecov` for LRG should include an amplitude-mode correction calibrated on the holi mocks (two rank-one terms, $\sigma_{A_0}^2 \bar{P} \bar{P}^T$ and $\sigma_{A_2}^2 \bar{P}_2 \bar{P}_2^T$) and the Gaussian-like rescaling $1 + a$.
6. Next steps: a super-sample module (isotropic and tidal responses in redshift space, local average for the realised normalisation), validated against the amplitude scatter measured here; the parameter test with the DESI model derivatives; the same analysis for the other tracer bins.